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How to design the fair experimental classifier evaluation

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ABSTRACT

Many researchers working on classification problems evaluate the quality of developed algorithms based on computer experiments. The conclusions drawn from them are usually supported by the statistical analysis and chosen experimental protocol. Statistical tests are widely used to confirm whether considered methods significantly outperform reference classifiers. Usually, the tests are applied to stratified datasets, which could raise the question of whether data folds used for classification are really randomly drawn and how the statistical analysis supports robust conclusions. Unfortunately, some scientists do not realize the real meaning of the obtained results and overinterpret them. They do not see that inappropriate use of such analytical tools may lead them into a trap. This paper aims to show the commonly used experimental protocols' weaknesses and discuss if we really can trust in such evaluation methodology, if all presented evaluations are fair and if it is possible to manipulate the experimental results using well-known statistical evaluation methods. We will present that it is possible to choose only such results, confirming the experimenter's expectation. We will try to show what could be done to avoid such likely unethical behavior. At the end of this work, we will formulate recommendations on improving an experimental protocol to design fair experimental classifier evaluation.

1. Introduction

Supervised classifier learning means that a classifier *learned* on the basis of a training dataset of labeled examples. Formally, a classifier Ψ is a function mapping a feature space $\mathcal{X}$ into a set of predefined class labels $\mathcal{M}$

$$\Psi : \mathcal{X} \rightarrow \mathcal{M} \tag{1}$$

There are many *learning algorithms* (see, for example, [1]) to infer that function from a training dataset. A learning algorithm generalizes from a training data to *unseen* situations, thus enabling to determine class labels for new objects outside from training data. To answer the question about the *goodness* of a new, designed classifier, we have to propose a robust method to evaluate it.

A plethora of scientific articles proposing new classifiers reports their predictive performance based on computer experiments. Usually, the results are auspicious, and the authors conclude that their proposition can outperform the benchmark solutions. Nevertheless, it worth remaining that Wolpert's *no free lunch* theorem concludes that the best learning algorithm does not exist [2], so all conclusions drawn from experiments are conditioned in given datasets and a chosen experimental protocol. Despite this, the primary motivation of most of the researchers is to publish their work. However, most of the journals are not interested in negative results because they are not so attractive for the readers and seem not interesting. Therefore, many authors follow the rule *if you torture the data long enough, Nature will confess* formulated by Ronald Coase.[1] It means that even if we design an experiment of the classifier evaluation according to the guides on classifier testing as [3–5], then there is still room for making mistakes or cheating. Franelli [6] has published an interesting study that reports that ca. 2% of scientists admitted to manipulating data or experiments. Simultaneously, they also noticed that this is a very popular behavior among the scientific community.

Of course, the most desirable approach is the data-fee one, based on an analytical evaluation of a predictive model, i.e., by the formulation of an appropriate theorem that shows the quality of the proposed method. Unfortunately, it is impossible for most of the methods, and the driven conclusions are based on strong and unpractical assumptions.

The problem of evaluating a classifier performance is tackled by using the metric to summarize the specific conditions of

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interest. The list of supervised classification metrics may be very large and depends on the problem under consideration. On the one hand, we may focus on standard metrics, and those designed *ad-hoc* for specific classification problems can be found, for example, in the well-known handbooks on machine learning, or pattern classification as [7–9]. On the other hand, it is also worth noticing that some authors criticize the use of the scalar classifier performance measures in favor of graphical methods [10], which are seen as a better choice to capture the complexity of a classifier evaluation process. While designing a classifier, we would like to estimate the proposed classifier's predictive performance and know if it performs better than a competitor or a set of classifiers for a given problem [11]. Some statistical inference is required (statistical tests or confidence intervals) to state that the observed difference in performances is not simply due to a chance (i.e., it is statistically significant). This paper follows the well established and most commonly used inference framework based on the frequentist statistics. One has, of course, to mention that there are some recent works based on the different framework, the Bayesian inference one [12], but we do not follow it. It is now generally agreed that the classifier evaluation process should include the following steps [3,5,13–16]:

1. selecting the appropriate metric(s) to characterize classifier performance,
2. selecting the pool of benchmark datasets,
3. choosing an appropriate metric estimation method,
4. significance difference testing using recommended statistical tests,
5. running the evaluation.

Due to the complexity of machine learning methods, we must learn about their behavior using experiments. The results of a single experiment are probabilistic, subject to variance in data, or a model. Controlling for these sources of variance requires appropriately using advanced statistical methods. This could be a challenge for many practitioners and scientists in machine learning who are not familiar with the theoretical foundations of statistical inference and must learn some rigor to use properly off-the-shelf tools.

Considering the above, our paper is primarily targeted to the Authors and Readers whose research is related to machine learning to give them practical recommendations for the correct use of the required statistical methodology in experimentation during the evaluation of their algorithms. In this work, we also show the proper interpretation of the obtained statistical results and what is necessary for this — some background knowledge on significance testing to understand the statistical methods used.

In addition to the above stated, crucial educational purpose, our paper's main contribution is to formulate newly revised recommendations related to using statistical tests during experimentation and evaluation procedures. In particular, based on the conducted, massive experiments, we have proved that the number of possible partitionings supporting false conclusions is smaller than in the case of uncorrected T-test. Nobody has done such simulations before. This enabled us to the postulation of validated recommendations for comparing two classifiers on one dataset experimentally. We also propose new recommendations for comparing two and more classifiers on many datasets enabling "bypassing" undesirable properties of some base methods.

We want to point to the pattern classification community to be cautious when we conclude the classifier quality based on experiments. We also would like to warn the research about using appropriately designed experiments on classifier comparison methods. We also consider the term of *replicable research* to answer the question if the reported experiments' presentations are always fair.

This paper starts with a short description of the most critical aspects of classifier evaluation steps [17]. Then we present two specially designed original experiments to illustrate problems arising from the improper use of statistical methods. The first experiment shows how inappropriate selection of folds in the popular *k*-fold cross-validation method used for classifier performance metric estimation may lead to classifier behavior instability. Consequently, it could lead to entirely different results (so-called *data shift problem*, discussed later). The second experiment presents problems with the commonly used statistical test, namely the *Friedman test* when comparing multiple classifiers on multiple datasets. We show how changing the pool of classifiers being compared influences the results. The next part includes new recommendations for the classifier evaluation, which are followed by conclusions.

Table 1
Confusion matrix for a two-class problem.

	PREDICTED POSITIVE	PREDICTED NEGATIVE
POSITIVE CLASS	*True Positive (TP)*	*False Negative (FN)*
NEGATIVE CLASS	*False Positive (FP)*	*True Negative (TN)*

2. An overview of classifier evaluation steps

This section aims to take a closer look at the five-step classifier evaluation methodology stated in the Introduction. In the beginning, we should underline that according to the Bayes decision theory [7], the best metric is the overall risk. However, using this metric requires knowledge of the cost matrix (also known as loss function), which expresses the cost of making a mistake between classes. Unfortunately, the information on the error cost should be obtained from the user who, in many cases, cannot estimate it. One of the interesting trends is finding mechanisms to obtain such knowledge from the user as the fuzzy approach [18] and the utility-based approach [19].

2.1. Selecting classifier metrics to characterize performance

Many metrics for classifier evaluation have been proposed (for interesting taxonomy, e.g., Japkowicz and Shah [14]). In many cases, metrics are determined on the basis of *confusion matrix*, which summarizes the number of instances from each class classified correctly, or incorrectly as the remaining classes [8]. Let us consider first the two-class classification task. The most popular metrics measuring classifier performance are the *Accuracy* (*Acc*) and the *Classification error* (*Err*). These can be computed from a 2×2 confusion matrix (see Table 1) as:

$$Acc = (TP + TN)/(TP + FN + TN + FP) \tag{2}$$

$$Err = (FP + FN)/(TP + FN + TN + FP) \tag{3}$$

Often we are interested in classifier evaluation on only a part of the data, i.e, positive or negative data. *True Positive Rate* (*TPR*) also known as *Recall* or *Sensitivity*), *True Negative Rate* (*TNR*), and *Precision* are the examples:

$$TPR = TP/(TP + FN) \tag{4}$$

$$TNR = TN/(TN + FP), \tag{5}$$

$$Precision = TP/(TP + FP) \tag{6}$$

Empirical evidence proves that accuracy and error rate are strongly biased to favor the majority class and might produce misleading conclusions. This fact motivated a search for new balanced measures obtaining a trade-off between positive and negative class performances.

Examples of such metrics are the arithmetic, geometric or harmonic means between the two components: *Recall* and *Specificity*. There are also other proposals trying to enhance one of the two components of the mean, for example *Index of Balanced Accuracy* [20]:

$$IBA_{\alpha} = (1 + \alpha(TPR - TNR)) \times TPR \times TNR \quad (7)$$

and $F_{\beta}score$ [21]:

$$F_{\beta}score = \frac{(\beta^2 + 1) \times Precision \times Recall}{\beta^2 \times Precision + Recall} \quad (8)$$

The parameters α, β can be tuned to obtain different trade-offs between both components. Nevertheless, using such metrics could be dangerous because their parameter α and β should set properly. Brzezinski et al. [22] show, e.g., that inappropriate parameter setting for $F_{\beta}score$ may favor the majority class for imbalanced data classification task. The interesting study of metrics behaviors for the chosen classification task could also be found in [23].

When misclassifications should be quantified in terms of costs, the *cost matrix* can be used to weight the entries in the confusion matrix. When the classification costs are difficult to obtain, the above mentioned balanced metrics may be used to set more relevance to the costliest misclassification. The *ROC* (*Receiver Operating Characteristic*) curve [24] has also been widely used in this case. Nevertheless, recent studies have shown that *AUC* (*Area under the* ROC *curve*) is a fundamentally incoherent measure (see for example [25], where the new alternative, the *H* measure has been proposed).

Performance measures for multi-class classification are still an open problem. Two methods are commonly used:

- *Macroaveraging* (per category) takes the average of measures on separate classes:

$$B_{macro} = \frac{1}{M}\sum_{i=1}^{M} B(TP_i, FP_i, FN_i, TN_i) \quad (9)$$

  where B is a binary metric, M stands for number of classes, while TP_i, FP_i, FN_i, TN_i are counted for binary task using OVA (*One-Versus-All*) decomposition and i denote the positive class.
- *Microaveraging* (per case) sums up individual TP_i, FP_i, FN_i, TN_i for different classes to get a measure:

$$B_{micro} = B\left(\sum_{i=1}^{M} TP_i, \sum_{i=1}^{M} FP_i, \sum_{i=1}^{M} FN_i, \sum_{i=1}^{M} TN_i\right) \quad (10)$$

In this paper, we focus on the metrics since they are a popular way to measure classification quality. However, these measures do not capture all the information about the quality of classification methods. Thus graphical methods may also be recommended. The presented list of metrics is by no means exhaustive, but it is worth mentioning the work by Branco et al. [26], where authors reported the following metrics which may be used in multi-class imbalanced data classification task: *Average Accuracy* (*AvAcc*), *Class Balance Accuracy* (*CBA*), *multi-class G-measure* (*mGM*), and *Confusion Entropy* (*CEN*) [27]. They are expressed as follows:

$$AvAcc = \frac{\sum_{i=1}^{M} Recall_i}{M} \quad (11)$$

where $Recall_i$ is counted using TP_i and FN_i,

$$CBA = \frac{\sum_{i=1}^{M} \frac{mat_{i,i}}{max\left(\sum_{j=1}^{M} mat_{i,j}, \sum_{i=1}^{M} mat_{j,i}\right)}}{M} \quad (12)$$

where $mat_{i,j}$ stands for the number of instances of the true class i that were predicted as class j.

$$mGM = \sqrt[M]{\prod_{i=1}^{M} Recall_i} \quad (13)$$

$$CEN = \sum_{i=1}^{M} P_i \cdot CEN_i, \quad (14)$$

where

$$P_i = \frac{\sum_{j=1}^{M} mat_{i,j} + mat_{j,i}}{2 \cdot \sum_{i,l=1}^{M} mat_{k,l}}$$

and

$$CEN_i = -\sum_{j=1, i\neq j}^{M} \left(P_{i,j}^{i} \log_{2(M-1)}(P_{i,j}^{i}) + P_{j,i}^{i} \log_{2(M-1)}(P_{j,i}^{i})\right)$$

additionally,

$$P_{i,i}^{i} = \begin{cases} 0 & ij \\ \frac{mat_{i,j}}{\sum_{j=1}^{M} mat_{i,j} + mat_{j,i}} & i \neq j \end{cases}$$

There are other vital aspects of classification, such as robustness to noise, scalability, stability after concept drift [28], and computational and memory complexities, which we do not address here.

2.2. Metric estimation using appropriate method

After selecting the appropriate classifier performance metric, we would like to know its true value, which means how a learned classifier behaves on a whole domain. We aim to use an appropriate estimation method to approximate metric value and based on the selected examples from the domain (i.e., a training dataset). We present estimation methods in the case of *classification error*, but they can also be used to estimate any chosen metric.

In statistics, the quality of an estimator is usually assessed using *mean squared error* (*MSE*) [29], which is the sum of the variance of the estimator and its squared bias. The *bias of* an estimator is a difference between its expected value and the true value of the parameter (here a metric) being estimated. The widely known *bias–variance dilemma* [30] is the conflict in minimizing these two sources of error simultaneously. We are interested in the estimation method with a small bias, which guarantees that the estimated metric value is close to the true unknown value. Similarly, a low variance of the estimation method is also desirable as high variance results in unstable and not reproducible metric approximations. What, in turn, may lead to inconsistent conclusions when comparing classifiers.

This section first describes the most popular standard estimation methods, only shortly, as they are well known in the machine learning community. Then, we present some selected methods to improve those standard ones in terms of bias or variance.

The *hold-out* method of metric estimation divides the available dataset randomly into mutually exclusive training and testing subsets, which are then used for classifier learning and metric estimation, respectively. The larger the training subset, the smaller the bias, but at the same time, the larger the variance. The desired bias–variance trade-off can be obtained by the appropriate proportion of train to test examples (usually 2/3 to 1/3). This method gives a pessimistically biased metric estimator. Resampling methods like cross-validation or bootstrap can obtain better metric estimators.

The k-fold cross-validation divides a dataset into k equal-size and mutually exclusive folds, of which (k-1) are used for training and a remaining one for metric estimation. This procedure is repeated k times, i.e., until all the folds are used for metric estimation. As a result, we obtain k different models and k values of metric estimators. The final metric value, here the classification error is estimated as the average of those k values of metric estimators. This method reduces the bias compared with a *hold-out* method, but increasing a number k of folds also increases the metric estimator's variance. Usually, the k should depend on the dataset size, i.e., the smaller size, the bigger k. The *k=10* is recommended, but *k=5* is also popular.

We should note that in k-fold cross-validation, through high overlapping in the training folds, the main independence assumption of many statistical tests used further for statistical comparison is not fulfilled. This can strongly affect the bias of the metric estimator.

The *leave-one-out* is the degenerate case of k-fold cross-validation where $k = N$ (where N is the size of the dataset), i.e. only one example in test set for metric estimation. This results in the quasi-unbiased metric estimator, but at the same time, largely increases the variance. In statistics, *bootstrapping* [31] relies to any method using random sampling with replacement. There are many variants of applying bootstrapping to the problem of classification error estimation (see, for example [32]). One example is the following, so-called *leave-one-out bootstrap* estimate. Each data example from the original dataset is evaluated only using the classifiers learned on those bootstrap samples that do not contain that example. However, this results in the pessimistically biased estimator of true classification error (e3) (the bias is similar to 2-folds cross-validation). In general, small bias and small variance of classification error (or any metric) estimators are desired. Below, we shortly present the selected techniques for bias and variance reduction.

The general idea behind bias reduction techniques is to subtract from the estimated metric (obtained using a particular estimator) the estimated, in some way, bias of that estimator. We recommend the well known in statistics *jackknife method* [33] for the bias estimation. The *jackknife estimator* of any parameter is found by systematically leaving out each example from a dataset and calculating the estimate, and then finding the average of these calculations. This technique can also be used to estimate the bias of any estimator. Based on the *jackknife method*, we can recommend simple bias reduction methods adapted for hold-out and cross-validation described in [34].

For the bias correction in bootstrap estimators of classification error, we recommend 0.632 bootstrap or 0.632+ bootstrap estimators [32]. Resampling methods can be used, e.g., by repeating several times the standard metric estimation method such as hold-out or cross-validation and averaging over all the results to reduce the variance of metric estimators. For example, the 10-repeated 10-folds cross-validation [35] presents a good bias–variance trade-off.

However, random splitting into the train and the test subsets performed in hold-out and cross-validation may lead to the partitions (folds) that are not representatives of the original dataset (so-called *dataset shift problems* [36]. Dataset shift is a common problem when the joint distribution of inputs and outputs differs between training and test steps. When only the input distribution changes, we call it a *covariate shift*. *Dataset shift* can potentially result in the instability of the classifier behavior during comparison procedures, thus dramatically changing the contests' results.

In the next section, we illustrate by suitable experimental example, the *manifestation* of dataset shift resulting in a dramatic change in classifier comparison results. When *dataset shift* occurs, reducing the metric estimator's variance (for example, classification error) requires more repetitions of hold-out or cross-validation. A better way to reduce the *dataset shift* and the variance is the use of *stratified sampling* strategies [37] to create dataset partitions. For example, in *standard stratified cross-validation*, we maintain class distributions equal in all partitions (folds) by inserting the same number of instances of each class on each partition.

More refined schemes have also been proposed. For example, in *distribution-balanced stratified cross-validation* [37], we keep the underlying data distribution by maximizing diversity on each fold while maintaining the similarity between folds. When datasets are small, e.g., in such domains as medicine, we should take special precautions. In such situations, the limited number of examples increases the metric estimator's variance, thus making the comparison results useless. In the next section on selecting datasets, we mention how to detect such situations and preclude such uncertain metric estimates.

2.3. Selecting the pool of datasets

The *No free lunch* theorem [38] in machine learning states that for any two classifiers, there are as many classification problems for which the first classifier outperforms the second as vice versa. Roughly speaking, no one model works best for all possible situations.

Therefore, it is a waste of time trying to show that one classifier is, on average, better than the others. Instead, we should investigate the conditions of the important domain problems at which our classifier behaves better or worse than others. These specific domain conditions can be effectively *mitigated* and explored using artificially generated datasets. Based on such mentioned explorations, we should carefully select the pool of *representative* datasets for classifier evaluation. Unfortunately, many of the current classifier contests employ the standard benchmark datasets that might not represent the domain problems at hand. Thus, in many cases, the representation assumption is questionable. The domain problems should guide the choice of datasets.

As mentioned in the previous section, we should also keep track of when the dataset is too small to prevent uncertain metric estimates. In such cases, the limited number of examples in the dataset is not enough good representation of the classification problem at hand. In literature, we may find some interesting solutions. Here, we mention only the simple, and in our opinion, the most interesting solution based on permutation tests [39].

The authors introduced the so-called *permutation-based p-value*, which counts the situations in which the classifier learned on the original dataset's permutation is better than on the original. Each of the k permutations is obtained by re-ordering class labels in the original dataset. Thus, large empirical *p values* suggest that the metric estimates are uncertain.

It is worth mentioning that during the experimental evaluation, we should avoid toy or synthetic data. It is an inappropriate approach because it could favor the group of models whose biases better fit the known data distributions [9]. Such datasets could be used for illustrative examples, but they are inappropriate for the robust statistical analysis of the classifiers' predictive performance. Because of the lack of real datasets for some problems, one uses the real databases' permutations based on statistical analysis, which is also incorrect.

Table 2
Type I and II errors in hypothesis testing.

	H_0 is rejected	H_0 is accepted
H_0 is false	Correct	Type II error
H_0 is true	Type I error	Correct

2.4. Significance difference testing using recommended statistical tests

The null hypothesis *statistical tests* (ST) [29] are used extensively in machine learning to answer such questions like "does the proposed classifier performs better than a competitor or a group of some selected classifiers ?" Statistical tests enable us to answer whether the obtained difference in classifier performance is statistically significant.

At the same time, statistical tests have mainly been criticized (e.g., [40]), and we should be aware of their misuses and limitations, which, in turn, require their understanding. Therefore, we first shortly review the basics about ST and then describe the relevant ST that should be applied in different evaluation scenarios.

Let us assume the metric values obtained by two classifiers Ψ_1 and Ψ_2 are the realizations of two random variables $S1$ and $S2$. The so-called *null hypothesis* H_0 states that both classifiers have the same performance while the *alternative hypothesis* H_1 that they behave differently. The hypothesis testing procedure requires a *test statistic*, specific to a given test, which summarizes the information about the observed values of $S1$ and $S2$ in a new random variable with the known distribution.

The *p value* is calculated based on a random sample from a considered population to make the statistical decision whether or not to reject H_0. This *p value* is an estimate of the probability of obtaining the observed (or more extreme) values of the test statistic, given that H_0 is true. The statistical decision may be correct or incorrect. The probability of rejecting H_0 when it is true (type I error) is called a *significance level* of a statistical test and denoted as α.

If the observed *p value* is smaller than α, we assume that such a small *p value* is due to a wrong initial assumption and, therefore, H_0 is rejected. Then, we conclude that a result (here, the difference between performances of two classifiers being compared) is *statistically significant*. A summary of possible errors in testing hypotheses is provided in Table 2.

However, the information provided by the *p value* does not tell us how probable H_0 is, but about how probable it is to observe the obtained, or more extreme values of a test statistic when H_0 is true [41]. If the obtained *p value* is small, we decide to reject H_0.

In comparing k classifiers, we assume the observed metric values are the realizations of $S_1, \ldots, S_k$ random variables. The null hypothesis H_0 states that all the classifiers show the same performance, while the alternative, that some of them have different.

After rejecting H_0, we are interested in detecting which specific classifiers have different performance. This is obtained by *post-hoc tests*, which involve $k(k-1)/2$ pairwise comparisons. When performing multiple comparisons (i.e. *multiple hypothesis testing*), one has to control the *family-wise error*, i.e., the probability of making at least one *type I error* in any of the comparisons [42]. The family-wise error can be controlled by correcting the α value in the pairwise comparisons.

The simplest and more conservative is the *Bonferroni correction* (i.e., dividing the significance level by the number of tests). More powerful techniques are also available in the literature to adjust the significance level α, for example, the *Holm, Hochberg* or *Finner* adjustments [43]. However, some alternatives to statistical tests have been proposed in the literature (for example, confidence intervals [44] or Bayesian methods [45]), currently, there is no clear solution to substitute statistical tests.

Below we shortly describe the relevant STs that should be applied in three evaluation scenarios, i.e., for a comparison of:

- two classifiers on one dataset,
- two classifiers on multiple datasets,
- multiple classifiers on multiple datasets.

For the comparison of two classifiers on one dataset, several parametric tests have been proposed. As they assume that data follow the normal distribution, this should be checked by appropriate tests of normality. Moreover, through high overlapping in the training folds during metric estimation (e.g., in the commonly used cross-validation), the independence assumption of those parametric statistical tests is violated, which can strongly affect the bias as the variance of metric estimators. Therefore, the corrected versions of classical statistical tests have been proposed.

In ith of the m repetitions (iterations) of the repeated estimation such as hold-out, repeated cross-validation, etc., a random data partition is conducted and the values for the metrics $A^{(i)}_{\Psi 1}$ and $A^{(i)}_{\Psi_2}$ of compared classifiers Ψ_1 and Ψ_2, are obtained. Unfortunately, most of the works employ the classical *t test* based on the T-statistic:

$$T = \frac{\bar{A}}{\sqrt{\left(\frac{1}{m}\right) \cdot \sum_{i=1}^{m} \frac{(A^{(i)}-\bar{A})^2}{m-1}}} \sim t_{m-1} \tag{15}$$

where $\bar{A} = \frac{1}{m}\sum_{i=1}^{m} A^{(i)}$ and $A^{(i)} = \left(A^{(i)}_{\Psi_1} - A^{(i)}_{\Psi_2}\right)$. This test should not be used due to its well-known limitations when comparing two classifiers.

In cross-validation, due to high overlapping of training folds in successive iterations, the corrected *t test* [35] has been proposed:

$$T = \frac{\bar{A}}{\sqrt{\left(\frac{1}{m(1-corr)}\right) \cdot \sum_{i=1}^{m} \frac{(\bar{A}^{(i)}-\bar{A})^2}{m-1}}} \sim t_{m-1} \tag{16}$$

where a parameter *corr* is unknown and cannot be unbiasedly estimated. For the repeated cross-validation, we recommend to use the following corrected *t test* [35]:

$$T = \frac{\bar{A}}{\sqrt{\left(\frac{1}{k\cdot m} + \frac{N_{ts}}{N_{tr}}\right) \cdot \sum_{i=1}^{m} \sum_{j=1}^{k} \frac{(A^{(i,j)}-\bar{A})^2}{k\cdot m-1}}} \sim t_{k\cdot m-1} \tag{17}$$

and N_{ts}, N_{tr} are the number of samples in the test and train partitions and $A^{(j,i)}$ is the metric difference on jth fold and the ith repetition.

McNemar's test [43] is a non-parametric alternative for comparing two classifiers. This test is based on the hold-out estimation method. However, it does not take the variability due to a choice of dataset or a learning algorithm's internal randomness.

For the comparison of two classifiers on multiple datasets the Wilcoxon signed-ranks test [43] is widely recommended. It ranks the differences

$$d_i = A^{(i)}_{\Psi_1} - A^{(i)}_{\Psi_2} \tag{18}$$

between metrics of two classifiers $\Psi 1$ and Ψ_2 obtained on ith of N datasets, ignoring the signs. The test statistic of this test is:

$$T = \min(R^+, R^-) \tag{19}$$

where:

$$\begin{aligned} R^+ &= \sum_{d_i>0} rank(d_i) + \frac{1}{2}\sum_{d_i=0} rank(d_i) \\ R^- &= \sum_{d_i<0} rank(d_i) + \frac{1}{2}\sum_{d_i=0} rank(d_i) \end{aligned} \tag{20}$$

Table 3
scikit-learn classifiers selected for experimental evaluation.

ACCR.	CLASSIFIER	PARAMETERS
GNB	*Gaussian Naive Bayes*	Without prior probabilities and with *variance smoothing* at $1e-9$.
KNN	*k-Nearest Neighbors*	With 5 neighbors, *uniform* weights and *euclidean distance*.
CART	*Classification and Regression Tree*	With *gini* criterion, *best* splitter, minimal 2 samples to split and without limit of tree depth.

are the sums of ranks on which the Ψ_2 classifier outperforms Ψ_1, respectively. Ranks $d_i = 0$ are split evenly among the sums.

The recommended methodology for the comparison of multiple classifiers on multiple datasets is a two-step procedure. In the first step, the omnibus test is used to verify if at least one of the classifiers performs differently than the others. The Friedman non-parametric test [43] with Iman–Davenport extension is probably the most popular and commonly used omnibus test. Let R_{ij} be the rank of the jth of K classifiers on the ith of N data sets and

$$R_j = \frac{1}{N}\sum_{i=1}^{N} R_{ij} \tag{21}$$

is the mean rank of the jth classifier. The test compares the mean ranks of the classifiers and is based on the following test statistic:

$$\begin{aligned} F_F &= \frac{(N-1)\chi_F^2}{N(K-1)-\chi_F^2} \\ \chi_F^2 &= \frac{12N}{K(K+1)}\left[\sum_{j=1}^{K} R_j^2 - \frac{K(K+1)^2}{4}\right] \end{aligned} \tag{22}$$

which follows an F distribution with $(K-1)$ and $(K-1)(N-1)$ degrees of freedom.

If we reject null hypothesis, then, in the second step *post-hoc analysis* with multiple hypothesis testing is conducted. For the described above Friedman test, the *post-hoc test statistic* comparing the rth and sth classifiers is based on the mean ranks and has the form:

$$z = \frac{R_r - R_s}{\sqrt{\frac{K(K+1)}{6N}}} \tag{23}$$

The family-wise error rate can be controlled through the mentioned above adjustments.

When the number of classifiers to be compared is less than five, other ranking techniques like Friedman aligned ranks [43] or Quade test [46] are more useful. Nevertheless, they will obtain better results, but only when the experimental study settings are proper. Recently, the new, weighted ranking methods [47] have been proposed as a more robust alternative. Nevertheless, many experiments with different settings are required to test their reliability and ranking fairness.

3. Experimental studies illustrating improper use of evaluation methodology

This section aims to show the chosen flaws of commonly used experimental protocols, i.e., how their improper use can dramatically change the results of classifier comparison. In the first experiment, we will focus on the dataset shift problem in cross-validation, while in the second, we show the problems with mean-ranks in the Friedman test.

Table 4
Overview of datasets applied in tests for Section 3.1.

DATASET	SAMPLES	FEATURES	CLASSES
breastcan	683	9	2
wisconsin	699	9	2
ionosphere	351	34	2
sonar	208	60	2
balance	625	4	3
monkone	556	6	2
heart	270	13	2
liver	345	6	2
hayes	160	4	3
german	1000	24	2
iris	150	4	3
monktwo	601	6	2
wine	178	13	3
yeast3	1484	8	2
monkthree	554	6	2

3.1. Problems with dataset shift in cross-validation

Let us describe the details of the conducted experimental study to illustrate how partitioning methods in cross-validation, the commonly used metric estimation method, introduce dataset shift (here, strictly the covariate shift) and show its impact on the classifier comparison results.

We want to show that:

Depending on the choice of the dataset folds in cross-validation, it is possible using the same dataset and the same two classifiers and the same significance test that one classifier outperforms the other and that there is not enough evidence for this.

In other words, we would like to show that if we have the sufficiently computational power to repeat random dataset divisions, we will lead to a situation enabling us to reasonably support three different hypotheses about the relationship between the accuracies of the compared classifiers, i.e., that one classifier outperforms the other or vice versa or that there is not a statistically significant difference between them. It must be clearly emphasized that such a practice is not consistent with the theory and normal practice of hypothesis testing, but we would like only to show that such extreme cases could happen.

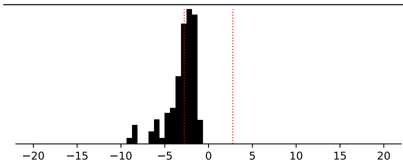
The specially designed experiment described below illustrates the mentioned procedure leading to the support of different hypotheses depending on the particular random division.

Eighteen popular datasets from UCI *Machine Learning Repository* (Table 4) [48] and three classifiers (Table 3) were adopted for the described experimental study.

For each dataset, 100 000 partitionings into 5 folds with standard stratification were performed, each being the basis for comparing the performance of 3 classifier pairs (constructed from the above mentioned). Thus, there were 54 different classifier comparisons (18 datasets with 3 pairs of classifiers). Classifier performance was measured by the earlier described *accuracy* metric (i.e., the ratio of classified adequately to all elements in a test set) estimated using standard stratified 5-fold cross-validation.

Table 5
Example summary of 100 000 repetitions of 5-fold cross-validation used to compare two classifiers — see the description in the text.

SONAR REGULAR



	GNB	KNN	T	p	D
≈	0.681	**0.784**	−3.06	0.04	–
	(0.07)	(0.06)			
=	0.667	0.726	−1.05	0.35	0.5418
	(0.07)	(0.06)			
−	0.653	**0.788**	−9.29	0.00	0.4582
	(0.07)	(0.06)			
+	–	–	–	–	0.0000

The classical *t test* for cross-validation was used, taking the significance level $\alpha = 0.05$ to assess the statistical significance of differences between the achieved classifier performances.

Each of 54 classifier comparisons was summarized in the separate table associated with a plot presenting the empirical distribution (histogram) of the values of *T-statistic* in the population of 100 000 partitionings (i.e., the random split into folds was repeated 100 000 times). The entries in such a table are as follows (see, for example, Table 5 for SONAR dataset depicted as the header). The first two columns include the accuracies of the compared classifiers (average or exact — see the description of the meaning of rows below), followed by the standard deviations beneath, the mean value of *T-statistic*, the corresponding *p-value* and the percentage of partitionings (in a pool of 100 000) in which the situation described in the line occurred (D). The meaning of the values in the following rows in each table are as follows:

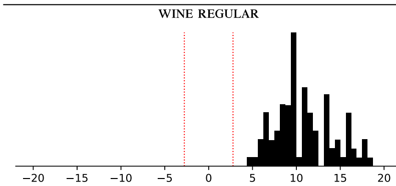
- ≈ the average values of classifier accuracy and *T-statistic* in 100 000 repetitions,
- = the values of the classifier accuracy with the *T-statistic* value nearest to zero, contained outside the critical region, symbolizing the lack of significant differences between the accuracies of the compared classifiers,
- − the values of the classifier accuracy with the lowest value of *T-statistic* inside the critical region, symbolizing a statistically significant advantage of the classifier from the right column,
- + the values of the classifier accuracy with the highest value of *T-statistic* inside the critical region, symbolizing a statistically significant advantage of the classifier from the left column.

Therefore, the interpretation of the example table will be as follows. The average value of the observation indicates a predominance of KNN algorithm with the accuracies of approximately 78% for KNN and 68% for GNB. The average *T-statistic* value is −3.06, which gives *p value* at 0.04, which makes it a statistically significant difference. However, we may also observe that a significant difference between algorithms does not occur (line =), where *T-statistic* equals −1.05, which gives *p value* at 0.35. Similar situations (no significant difference) occurs in 54% of the considered cases. Partitionings which show a significant advantage of the KNN algorithm occur in 46% of cases and the extreme value of *T-statistics* is −9.26, which is an outlier in the context of the problem under consideration.

In connection with the above observations, we can properly validate two contradictory hypotheses by pulling the dataset into folds adequately many times. Moreover, with the average value of *T-statistics* close to the test's critical value, approximately a half of the experiments will give us information about the KNN advantage, when all the other cases of partitionings will deny any statistical difference.

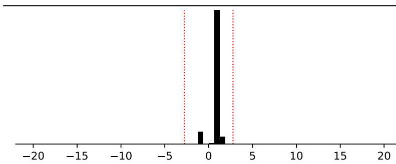
Table 6
Three example summaries of 100 000 repetitions of 5-fold cross-validation used to compare two classifiers showing **results supporting one hypothesis**.

WINE REGULAR



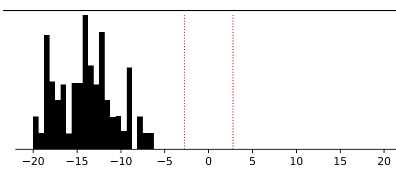
	GNB	KNN	T	p	D
≈	**0.974**	0.690	11.74	0.00	–
	(0.03)	(0.06)			
=	–	–	–	–	0.0000
−	–	–	–	–	0.0000
+	**0.977**	0.702	33.48	0.00	1.0000
	(0.03)	(0.06)			

SOYBEAN REGULAR



	GNB	CART	T	p	D
≈	0.998	0.989	0.87	0.43	–
	(0.01)	(0.04)			
=	0.978	0.978	0.00	1.00	0.4388
	(0.01)	(0.04)			
−	–	–	–	–	0.0000
+	–	–	–	–	0.0000

MONKONE REGULAR



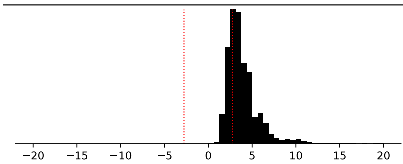
	GNB	KNN	T	p	D
≈	0.664	**0.945**	−18.24	0.00	–
	(0.04)	(0.02)			
=	–	–	–	–	0.0000
−	0.664	**0.957**	−47.89	0.00	1.0000
	(0.04)	(0.02)			
+	–	–	–	–	0.0000

The simplest of situations encountered in performed experiments is that each of a hundred thousand experiments leads to the same conclusion, as shown in the examples in Table 6. The distribution of computable *T-statistics* in such cases is either narrow enough to fit between the critical regions (SOYBEAN) or far enough from the critical value so that even outlier observations leading to other conclusions do not happen. It is worth noting, however, that even in such cases, outliers reveal a major deviation from the mean value (for example, an average of 11.74 in the WINE dataset and an outlier observation of 33.48). It is particularly important to note at this point that such unambiguous conclusions characterize only 12 of the 54 comparisons.

As many as 34 out of 54 comparisons, the dominant majority correspond to the situation presented in the example in Table 5,

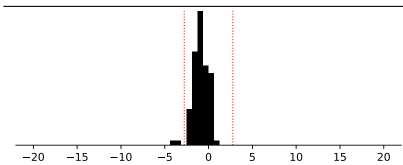
Table 7
Three example summaries of 100 000 repetitions of 5-fold cross-validation used to compare two classifiers showing results supporting **two different hypotheses**.

WINE REGULAR



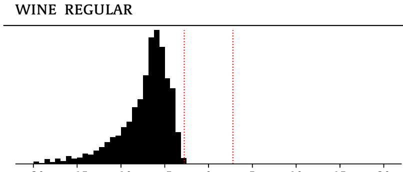
	GNB	CART	T	*p*	D
≈	**0.974** (0.03)	0.905 (0.05)	3.91	0.02	–
=	0.967 (0.03)	0.961 (0.05)	0.32	0.76	0.2751
−	–	–	–	–	0.0000
+	**0.972** (0.03)	0.944 (0.05)	69.20	0.00	0.7249

IRIS REGULAR



	GNB	KNN	T	*p*	D
≈	0.954 (0.03)	0.965 (0.03)	−1.00	0.37	–
=	0.960 (0.03)	0.960 (0.03)	0.00	1.00	0.9389
−	0.947 (0.03)	**0.973** (0.03)	−4.00	0.02	0.0204
+	–	–	–	–	0.0000

WINE REGULAR

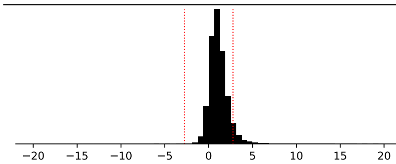


	KNN	CART	T	*p*	D
≈	0.690 (0.06)	**0.905** (0.05)	−7.48	0.00	–
=	0.724 (0.06)	0.847 (0.05)	−2.30	0.08	0.0011
−	0.702 (0.06)	**0.921** (0.05)	−57.30	0.00	0.9989
+	–	–	–	–	0.0000

where we can draw two contradictory hypotheses from the appropriate (let us emphasize — random) partitionings of patterns into folds. Examples here are presented in Table 7. In the case of the WINE dataset (comparing GNB and CART), we may observe a situation in which the GNB algorithm in averaging achieves an advantage over the CART. However, in a quarter of the cases, the random division of the dataset conclude that there are no significant differences. A much more interesting case is the IRIS dataset. There is no significant difference between the compared classifiers in 94% of divisions, but 2% of the partitionings leads to the conclusion that the KNN has a significant advantage over GNB. An even stronger example of this type is another case of the WINE dataset, this time comparing KNN and CART. Despite the predominance of the DTC over the KNN and the average difference in quality at 21%, we can still find one part-per-thousand of situations in which the statistical difference between them disappears.

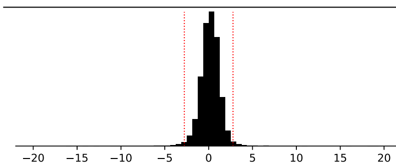
Table 8
Three example summaries of 100 000 repetitions of 5-fold cross-validation used to compare two classifiers showing results supporting **three hypotheses**.

LIVER REGULAR



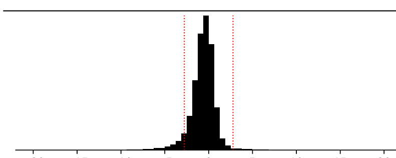
	KNN	CART	T	*p*	D
≈	**0.663** (0.05)	0.634 (0.05)	$2.6{*}10^{10}$	0.00	–
=	0.643 (0.05)	0.643 (0.05)	0.00	1.00	0.9413
−	0.638 (0.05)	**0.655** (0.05)	−6.00	0.00	0.0003
+	**0.699** (0.05)	0.641 (0.05)	$2.6{*}10^{15}$	0.00	0.0584

GERMAN REGULAR



	KNN	CART	T	*p*	D
≈	0.691 (0.02)	0.688 (0.03)	0.15	0.89	–
=	0.688 (0.02)	0.688 (0.03)	0.00	1.00	0.9807
−	0.693 (0.02)	**0.709** (0.03)	−16.00	0.00	0.0096
+	**0.712** (0.02)	0.682 (0.03)	18.97	0.00	0.0097

DIABETES REGULAR



	KNN	CART	T	*p*	D
≈	0.691 (0.03)	0.701 (0.04)	−0.67	0.54	–
=	0.682 (0.03)	0.682 (0.04)	0.00	1.00	0.9286
−	0.681 (0.03)	**0.716** (0.04)	−21.90	0.00	0.0610
+	**0.714** (0.03)	0.677 (0.04)	11.06	0.00	0.0104

Table 9
The number of supported hypothesis among 54 cases depending on the metric used to calculate the T-test.

TEST USED	EQ.	NUMBER OF SUPPORTED CONCLUSIONS		
		1	2	3
classical	*15*	12	34	8
corrected (corr=.1)	*16*	11	35	8
corrected (corr=.2)	*16*	11	35	8
corrected (corr=.3)	*16*	9	39	6
corrected (corr=.4)	*16*	9	39	6
corrected (corr=.5)	*16*	9	39	6
corrected (corr=.6)	*16*	13	36	5
corrected repeated CV	*17*	23	31	0

However, the most interesting is the third, last group of observations, which consists of 8 out of 54 examples (Table 8). There

Table 10
R classifiers selected for experimental evaluation.

ACCR.	CLASSIFIER	R PACKAGE	PARAMETERS
GBM	*Gradient Boosting Machine*	`gbm`	With *bernoulli* distribution, 100 trees, additive model, minimum 10 samples to create node and 50% of patterns selected to propose next tree in the expansion.
KNN	*k-Nearest Neighbors*	`kknn`	With 7 neighbors and euclidean distance.
GLM	*Generalized Linear Model*	`glmnet`	With lasso regularization, 100 lambdas and standardization.
DT	*Recursive Partitioning And Regression Trees*	`rpart`	Using *class* mode.
RF	*Classification And Regression With Random Forest*	`randomForest`	With 500 trees, sampling with replacement, without node limit and with normalized votes.
RF1	*Ranger*	`ranger`	With 500 trees, regularization factor 1 and square root of samples necessary to split a node.
LDA	*Linear Discriminant Analysis*	`MASS`	With training prior probabilities, *moment* method and $toi = 1e-4$.
NN	*Single-Hidden-Layer NN*	`nnet`	With proportional weights, maximum 100 iterations, BFGS optimizer and least-square entropy.
SVM	*Support Vector Machines*	`e1071`	With linear kernel and $\epsilon = 0.1$.

are cases, depending on which of the random partitionings we select for the experiment, we may get the validation of each possible conclusion. The LIVER dataset is particularly interesting here, where only approximately 30 out of 100 000 cases show the CART algorithm superiority over KNN, with approximately 94% of their equal quality and 5% of KNN advantage cases.

We also performed the above-described simulation using the remaining two corrected versions of *T-statistics* described in the previous section, the first with different parameter *corr*, the second — the corrected for repeated cross-validation. Table 9 presents a summary of the set of cases described in this subsection for the classical *t-test* and its corrected versions (introduced in Section 2.4), applied for calculations of T-statistic. For the corrected repeated cv - the 2 × 5-fold cross-validation has been used.

The application of tests with parameter correction (*corrected*, Eq. (16)) leads to a gradual reduction of cases in which it is possible to find support for three different conclusions, but for cases in which the result of all comparisons is evident and only leads to one conclusion. The overall trend is that the percentage of situations in which two conclusions appear (the lack of statistical differences or the predominance of one of the classifiers) is strengthened. Elimination of cases in which observations made from statistical analysis lead to three different conclusions occurs only in the case of a t-test for (*corrected repeated* cv, Eq. (17)), in which the most leading situations occur, among all divisions, to a clear conclusion.

Both the source code used to conduct the described experiment and a report with all the results is available in the public Git repository.[2]

3.2. Problems with mean-ranks in Friedman test

In this part, we will describe the details of the conducted experimental study to illustrate the fundamental problem with mean-ranks in the Friedman test, which is commonly used to compare multiple classifiers on multiple datasets in different contests.

We want to show that:

Depending on the choice of the pool of classifiers for the comparison, one can change the result of comparing the performances of the particular two classifiers.

[2] https://github.com/w4k2/lie

Table 11
Characteristics of datasets from UCI Machine Learning Repository used in the experiments.

DATASET	INSTANCES	FEATURES	CLASSES
Australian Credit Approval	690	14	2
Banknote authentication	1372	4	2
Breast Cancer	268	9	2
Ecoli	336	7	8
German Credit Data	1000	20	2
Glass	214	9	6
Haberman's Survival	306	3	2
Ionosphere	351	34	2
Iris	150	4	3
Liver Disorders	345	6	2
MONK 1	432	6	2
MONK 2	432	6	2
MONK 3	432	6	2
PIMA Indian Diabetes	768	8	2

Let us first consider the pool composed of the 9 classifiers (utilized as wrappers from R packages in scikit-learn environment) described in Table 10.

The experiments were carried out using 14 datasets from UCI MACHINE LEARNING REPOSITORY [48], whose characteristics are presented in Table 11.

The classifiers' error rates were estimated using 10-fold cross-validation and are reported in Table 12. As the *p value* of the Friedman test is very small ($p \approx 0.0003083$), we reject the null hypothesis about the same performances of 9 classifiers (on the significance level $\alpha = 0.05$). We can thus proceed with the post hoc tests to find differences among them. Assume for a moment that we are interested in comparison of classifiers RF with DT.

The *p values* (without adjustments) of the Friedman post-hoc test statistics are given in Table 13. We control the family-wise error rate through the Bonferroni correction (but the illustrated problem is valid regardless of the chosen correction). For the considered pair of classifiers the $pvalue = 0.06686$ is greater than the adjusted significance level

$$\alpha^* = \frac{0.05}{9 \cdot 8/2} = 0.001388$$

Thus, we cannot reject the null post-hoc hypothesis and conclude that performances of DT and RF classifiers *do not differ significantly*.

Let us compare now the mentioned two classifiers, RF and DT, in a pool of only 3 classifiers (i.e. with a classifier GLM). Based on a small $p \approx 0.0008793$ of Friedman test on these 3 classifiers, we reject the null hypothesis and proceed with the *post-hoc* analysis.

Table 12
Classifier errors calculated on different datasets from UCI machine learning repository using 10-fold CV.

	GBM	KNN	GLM	DT	RF	LDA	RF1	NN	SVM
1	0.4449	0.1623	0.1333	0.1478	0.1318	0.1420	0.1289	0.2028	0.1507
2	0.4446	0.0014	0.0218	0.0342	0.0065	0.0233	0.0094	0.0007	0.0000
3	0.3447	0.0300	0.0415	0.0472	0.0286	0.0401	0.0257	0.0529	0.0343
4	0.1369	0.0563	0.0741	0.1161	0.0592	0.0654	0.0742	0.0982	0.0565
5	0.3000	0.2900	0.2430	0.2540	0.2310	0.2310	0.2390	0.2940	0.2430
6	0.3839	0.2751	0.3504	0.3034	0.2049	0.3777	0.2142	0.6768	0.2798
7	0.2640	0.3194	0.2607	0.3030	0.2934	0.2540	0.3097	0.2739	0.2575
8	0.3592	0.1482	0.1169	0.1253	0.0654	0.1369	0.7400	0.1111	0.0598
9	0.0533	0.0046	0.0466	0.0733	0.0466	0.0200	0.0400	0.2133	0.0333
10	0.4200	0.3704	0.3134	0.3505	0.2663	0.3163	0.2780	0.3589	0.3241
11	0.2493	0.0993	0.3326	0.1755	0.0415	0.3326	0.0439	0.1989	0.0830
12	0.3288	0.1619	0.3311	0.2706	0.2547	0.3472	0.2315	0.3497	0.3219
13	0.3263	0.0866	0.2464	0.0136	0.0273	0.2259	0.0295	0.0820	0.0411
14	0.3488	0.2643	0.2251	0.2537	0.2368	0.2238	0.2355	0.3604	0.2278

Table 13
The *p values* (without adjustments) of the Friedman post-hoc test statistics.

	GBM	KNN	GLM	DT	RF	LDA	RF1	NN	SVM
GBM	–	–	–	–	–	–	–	–	–
KNN	0.12735	–	–	–	–	–	–	–	–
GLM	0.12735	1.00000	–	–	–	–	–	–	–
DT	0.49394	0.99892	0.99892	–	–	–	–	–	–
RF	0.00001	0.33479	0.33479	0.06686	–	–	–	–	–
LDA	0.04015	0.99998	0.99998	0.97381	0.61538	–	–	–	–
RF1	0.00013	0.63934	0.63934	0.20674	0.99996	0.87860	–	–	–
NN	0.97381	0.77327	0.77327	0.98882	0.00236	0.49394	0.01287	–	–
SVM	0.00021	0.70895	0.70895	0.25705	0.99979	0.91751	1.00000	0.01840	–

Table 14
The *p values* (without adjustments) of the Friedman post-hoc test statistics.

	DT	RF	GLM
DT	–	–	–
RF	0.00067	–	–
GLM	0.33187	0.06041	–

The *p values* (without adjustments) of the Friedman post-hoc test statistics are given in Table 14. Now, for the considered pair of classifiers the *p value*=0.00067 is less than an adjusted significance level

$$\alpha^* = \frac{0.05}{3 \cdot 2/2} = 0.0166$$

Therefore, we decide to reject the null post-hoc hypothesis and conclude now that performances of the two classifiers, DT and RF, *are significantly different*.

We see that the decision is conditional on the pool of classifiers under consideration. However, it is evident that the two classifiers' performances, DT and RF, should be the same in the above presented two cases.

If we adopt the Wilcoxon signed-rank test for comparing DT and RF classifiers separately, we obtain the *p value* $p \approx 0.00037$ which results in rejection of the null hypothesis. The conclusion is that RF and DT classifiers have significantly different performance: RF is significantly better than DT classifier (Table 13).

Because of the above illustrated fundamental problem with mean-ranks in the Friedman test, we discourage its use in the classifier evaluation procedure (i.e., when comparing multiple classifiers on multiple datasets). Instead, we suggest pairwise comparisons based on the Wilcoxon signed-rank test with a suitable correction to control the family-wise error rate. It is the most important difference between our suggested procedure and those described in [3] and [4].

4. Recommendations

Based on our solid literature studies and conducted massive computer experiments, we can formulate the following recommendations which, in our opinion, could help in selecting the right steps in classifier evaluation procedures.

- The most popular and frequently used performance measure (metric) for evaluating classifiers, the accuracy (or classification error) may be completely inappropriate when the datasets are imbalanced. The misclassification costs for false positives and false negatives are not the same (which is instead a typical situation). In such cases, the balanced metrics such as Index of Balanced Accuracy, F-score, geometric or harmonic means, or H measure are more appropriate. Moreover, we strongly recommend careful analysis of the chosen metric because of a given class' possible bias. *New performance measures that aim to obtain a trade-off between the evaluation of classification ability on both positive and negative classes need to be developed.*
- Using the parametric metrics as IBA_α or $F_{beta}score$ could lead to false conclusions because their incorrect parameter setting may favor one of the classes. Thus, *we should take special care in the use of parametric metrics, and we should set appropriate parameter values depending on the characteristics of a given decision problem.*
- Different classifier performance metrics reflect different insights into how a classifier performs. Therefore, multiple metrics need to be used to characterize the classifier's behavior. *Many research is required to investigate how to combine several metrics to yield a good summary of classifier characteristics.*
- In multi-class classification, imbalanced data is rather a rule and then new, specially designed metrics like *Average Accuracy* (AvAcc), *Class Balance Accuracy* (CBA), *multi-class G-measure* (mGM), and *Confusion Entropy* (CEN) should be used. Still, there is no full agreement among the researchers on which of the two techniques, macro or microaveraging is

a better choice. *Performance measures for multi-class classification are still the subject of intensive research that requires future in-depth empirical research*.

- We recommend using simple metrics because they are easier to interpret than aggregate metrics. The analysis of simple metrics should be supported by appropriate visualizations (e.q., radar chart) to help understand the relationship between the values of simple metrics for different classifiers.
- Regarding the estimation of a metric value: when the provided dataset is large enough, we recommend the usual hold-out, for moderate sizes — 10 repeated 10-fold cross-validation. When time is problematic, it is better to increase the number of repetitions than the number of folds. Additionally, we also recommend using simple techniques for bias and variance reduction like jackknife-based cross-validation or standard stratified cross-validation (mainly when *dataset shift* occurs). In the case of very small datasets, it is essential to keep track when a dataset is not representative of the domain using, for example, permutation tests. *In general, estimation techniques enabling for better bias–variance trade-off should be developed*.
- According to *No free lunch theorem*, no model can work best for all possible tasks, and it is a waste of time trying to show that one classifier, on average, outperforms the others. Instead, *dataset selection should to guided by the specific domain requirements, as imbalance ratio, cost of feature acquisition or misclassification cost, dataset volume, needs of interpretability, application domain needs, etc. to identify specific conditions that cause the classifier to perform better than others*.
- When comparing two classifiers on one dataset, we recommend using *corrected T-test for repeated hold-out or repeated cross-validation*. Although the *dataset shift* problem also affects them, they take into account in their construction, on the one hand, the use of data division into folds, on the other hand, (as the experiments presented in the previous section have shown) the number of possible partitionings supporting false conclusions is smaller than in the case of uncorrected T-test.
- *We recommend using the Wilcoxon signed-rank test for two classifier comparison on multiple datasets*.
- When comparing multiple classifiers on multiple datasets, the commonly used Friedman test with the *post-hoc* tests, as based on the global mean-ranks, has a serious drawback — the result of the comparison is dependent on the pool classifiers considered. Because of this fundamental inconsistency, we discourage its use in machine learning. At present, we suggest instead to perform the multiple comparisons using the sign-test or the Wilcoxon signed-rank test with appropriate corrections for multiple hypothesis testing. Some alternatives for ranking have been proposed, but they still need more systematic studies to check their reliability and fairness. *Thus, the new, fair ranking approaches should be developed for the fair usage of Friedman test*.
- Last but not least, we have to realize that authors of classification algorithms should not test them because they are biased towards their high quality. Therefore, *the machine learning community should consider the possibility of building an independent classifier testing platform*, as environments used for data classification challenges (see, e.g., Kaggle).[3]

We want to emphasize that we were focusing on the canonical classification task and other types of classification tasks, as multi-label classification [49], or data stream classification [28], are not addressed here, since *they may impose specific conditions to the calculation of the metric*.

5. Conclusions

The presented work did not intend to provide an exhaustive framework related to statistical hypothesis testing methods based on the *null hypothesis significance testing procedure* for assessing the quality of predictive models, nor is a survey on this topic. The primary purpose of our article was to provide a better understanding of the overall classifier evaluation process for readers. Since there is no one specific procedure to evaluate the quality of classifiers, we hope that this work will help the machine learning community in which alternative methods should be chosen for a particular case. We also aimed to indicate the dangers that await researchers who employ statistical analysis tools automatically and to recommend how to design a fair experimental evaluation of the classifiers using statistical analysis.

Additionally, many of the researchers make wrong conclusions based on the statistical analysis, because they probably wrongly interpret the *p value* as the probability that the null hypothesis is true, that the sample result occurred by chance. The correct interpretation is that p value means that if the null hypothesis were true, a sample result of this extreme would occur only *p value* of the time [50]. Some of the researchers conclude that *p value* is not reliable measure, because of its sampling variability, what was named as p value *dance* by Geoff Cumming.[4] This observation is widely confirmed, e.g., by Simmons et al. [51]. On the other hand Nuzzo [52] observes that the replication of exact obtaining the same *p value = 1%* is 50%, not 99% as many could expect. Nevertheless, no matter which measure has been chosen, the crucial question is how to estimate its value reliably.

Our goal was also to show that currently applicable experimental protocols can be used for conscious or unconscious manipulation of the experimental results. The first case is the false hypothesis acceptance, which can lead to the publication of this type of results, because the widely accepted manner is the publication of positive results, rarely when scientific journal editors are interested in publishing negative results. On the other hand, the results published in the form of so-called *replicable research* may prefer the published solutions, e.g., by choosing a set of folds preferring such solutions.

Hence, either the recommendations formulated in our work should be used, or the statistical analysis should be abandoned altogether, which was also done, e.g., by *Basic and Applied Social Psychology*. The editors of this journal banned the use of *null hypothesis significance testing procedure* and related statistical procedures from their journal.[5]

CRediT authorship contribution statement

Katarzyna Stapor: Conceptualization, Methodology, Formal analysis, Investigation, Validation, Writing - original draft, Writing - review & editing. **Paweł Ksieniewicz:** Visualization, Software, Conceptualization, Investigation, Writing - original draft, Writing - review & editing. **Salvador García:** Conceptualization. **Michał Woźniak:** Conceptualization, Methodology, Validation, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

3 https://www.kaggle.com/competitions

4 https://www.youtube.com/watch?v=ez4DgdurRPg

5 https://sciencebasedmedicine.org/psychology-journal-bans-significance-testing/

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